Accessible Rich Internet Applications (ARIA)

In short

Special labels and roles in code can tell screen readers what something is when HTML can't. Used badly, they can make a site much worse for these users. Use plain HTML first, and only add them when there's no other way.

What is it for?

Accessible Rich Internet Applications (ARIA) is a set of roles and attributes that make the web more accessible for users with disabilities. ARIA adds to HTML. It lets common web app interactions be passed to assistive technologies when there is no other way to do this.

There are some rules, set out by World Wide Web Consortium (W3C), about how ARIA should be used. They are as follows:

1. If a built-in HTML element or attribute already has the meaning and behaviour you need, use it. Don't take a different element and add an ARIA role, state, or property to make it accessible.
2. Do not change the built-in meaning of HTML elements unless you really have to.
3. All interactive ARIA controls must be usable with the keyboard.
4. Do not use `role="presentation"` or `aria-hidden="true"` on a focusable element. Using either of these on a focusable element will result in some users focusing on 'nothing'.
5. All interactive elements must have a name that assistive technology can read out, known as an accessible name (W3C definition).

Note: Not using ARIA is always better than using ARIA badly. Bad use of ARIA can make a website impossible to use for people who use assistive

technology. ARIA attributes should only be used when semantic HTML can't do the job.

The note below is adapted from MDN Web Docs: ARIA:

Note: There is a saying: "No ARIA is better than bad ARIA". In WebAIM's survey of over one million home pages, home pages with ARIA had more detected errors on average than those without it. ARIA is designed to make web pages more accessible, but if it's used incorrectly, it can make things worse.

How to support it

There are many different ARIA attributes. The two most commonly used ones are below.

aria-label and aria-labelledby

- Adds an accessible label to elements that do not have visible text. For example, a search button with only an icon should include an `aria-label` clarifying it.
- Use it on interactive elements, like `buttons`, and on landmarks, dialogs, tables, and elements with role="img". Don't use it on elements that have no role, like a plain div or span, or on text elements like p, because screen readers may ignore it there.

aria-hidden

- Hides elements from screen readers without removing them visually.
- This is ideal for decorative or redundant elements.

ARIA roles

ARIA roles add meaning to content. This lets assistive technologies (such as screen readers) present the content in the way users expect. These should be avoided when there is semantic HTML that can provide the same meaning (for example, don't use `role=listitem`, and instead use an `<li>`).

Useful links

- W3C - Rules of ARIA
- Mozilla: ARIA
- Accessibility essentials every front end developer should know
- List of ARIA attributes
- List of ARIA roles
- What I Wish Someone Told Me When I Was Getting Into ARIA

Abbreviations

The full form of each abbreviation used in this document.

ARIA: Accessible Rich Internet Applications

HTML: HyperText Markup Language

MDN: Mozilla Developer Network

W3C: World Wide Web Consortium